MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 7609

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode

See figure 1

Mounting position: Any

Weight: 4 ounces nominal

ABSOLUTE RATINGS: F1 = 150 MHz; F2 = 500 MHz

Parameter: / Unit:		Ef Vac See note 2	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Ehk Vdc	Ib mAdc	Pg1 W	Pg2 W	Pp W
Maximum:										
C Teleg:	(Up to F1)	---	2,000	-250	300	+150	250	2	12	250
C Teleg:	(F1 to F2)	---	1,250	-250	300	±150	250	2	12	250
TEST CONDITIONS:		26.5	2,000	Adjust	300	0	150	---	---	---

ABSOLUTE RATINGS: F1 = 150 MHz; F2 = 500 MHz

Parameter: / Unit:		T(Base seal) °C See note 1	T(Anode seal) °C See note 1	T(Anode core) °C See note 1	tk sec (min)
Maximum:					
C Teleg:	(Up to F1)	175	200	250	30
C Teleg:	(F1 to F2)	175	200	250	30
TEST CONDITIONS:		See note 3	---	---	120

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
Ⓓ 2212	Pulsing emission	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ik = 1.5 a; prr = 11 ±1; tp = 4500 µs min.; Ef = 23.8 Vac; tr = tf = 25 µs max; slope = 0.5% max; ripple = 0.1% max	0.65	II	Δik	---	200	ma
Ⓓ 1256	Electrode current (screen) (1)	Eb = 1,000 Vdc	0.65	II	Ic2	-3	+5	mAdc
Ⓓ 1256	Electrode current (screen) (2)	Eb = 200 Vdc; Ec2 = 150 Vdc; Ec1/Ib = 150 mAdc	0.65	II	Ic2	+5.0	---	mAdc
1261	Electrode voltage (grid)	Eb = 1,000 Vdc	0.65	II	Ec1	-32	-46	Vdc
1266	Total grid current	Ib = 125 mAdc See Note 5	0.65	II	Ic1	---	-15	µAdc
1372	Current division (method A, long pulse)	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ib = 1.0 a; prr = 11.0 +1.0; tp = 4,500 µs (min)	0.65	II	egk ic1 ic2	8 --- ---	18 250 260	v ma ma
1266	Primary grid emission (control)	Ic1 = 70 mAdc; t = 15; anode and screen grid floating	0.65	II	Isg1	---	-25	µAdc
1266	Primary grid emission (screen)	Ec1 = 0; Ic2 = 100 mAdc; t = 15; anode floating	0.65	II	Isg2	---	-250	µAdc
1301	Heater current		0.65	II	If	0.40	0.62	Aac
	Quality conformance inspection, part 2							
----	Power output (1)	F = 150 MHz; Ec1 = -90 Vdc; Ic1 = 25 mAdc max; Eg1/Ib = 250 mAdc	---	---	Po	225	---	W (useful
2214	Power output (2)	Class C amplifier; Ef = 24.3 Vac; F = 460 to 490 MHz; Eb = 1,250 Vdc; Ec1 = -90 Vdc; Ec2 = 250 to 300 Vdc; Ic1 = 20 mAdc max; Eg1/Ib = 250 mAdc See note 6	---	---	Po	145	---	W (useful
1316	Amplification factor	g1-g2; Ec2 = 300 Vdc; Ic2 = 50 mAdc; anode floating	---	---	Mu	4.0	6.0	---
1331	Direct-interelectrode capacitance	EIA standard shield No. 320 and No. 321, or equal	--- --- ---	--- --- ---	Cgp Cin Cout	--- 14.5 3.7	0.05 17.0 4.45	pF pF pF
1336	Heater-cathode leakage	Ehk = +150 Vdc; Ehk = -150 Vdc			Ihk Ihk	--- ---	150 150	µAdc µAdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality Conformance inspection, part 1							
----	Life-test (1) provisions	Group C; power output (1) t = 500 hours	---	---	---	---	---	---
----	Life-test (1) end points:							
Ⓓ 2212	Pulsing emission	Ef = 26.5 Vac	---	---	Δik	---	100	ma
1266	Primary grid emission (control)		---	---	Isg1	---	-100	µAdc
1266	Primary grid emission (screen)		---	---	Isg2	---	-250	µAdc
1336	Heater-cathode leakage	Ehk = +150 Vdc Ehk = -150 Vdc	--- ---	--- ---	Ihk Ihk	--- ---	150 150	µAdc µAdc
----	Life-test (2) provisions	Group C; Ec1 = Ec2 = Eb = 0; Ef = 29.1 Vac; t = 500 hours	---	---	---	---	---	---
----	Life-test (2) end point:							
1366	Interelement leakage resistance (cold)	supply voltage = 500 Vdc; Rs = 2.5 Meg g2 positive g2 positive g1 negative See note 7	--- --- ---	--- --- ---	Rg1g2 Rg2k Rg1k	10 10 10	--- --- ---	Meg Meg Meg
Ⓓ ----	Life-test (3) provisions	Group D; power output (2) t = 500 hours See note 9	---	---	---	---	---	---
----	Life-test (3) end points:							
Ⓓ 2212	Pulsing emission	Ef = 26.5 Vac	---	---	Δik	---	100	ma
1266	Primary grid emission (control)		---	---	Isg1	---	-100	µAdc
1266	Primary grid emission (screen)		---	---	Isg2	---	-250	µAdc
1336	Heater-cathode leakage	Ehk = +150 Vdc Ehk = -150 Vdc	--- ---	--- ---	Ihk Ihk	--- ---	150 150	µAdc µAdc
1155	Coolant pressure drop versus coolant flow (forced air)	No voltages See note 4 and 9	---	---	---	---	0.60	Inch of water
1042	Shock, specified pulse	No voltages; accel = 15G peak (min); D = 11 ±2 ms half-sine wave See note 8	---	---	---	---	---	---
---	Shock, end points:							
1261	Electrode voltage (grid)		---	---	Ec1	-32	-46	Vdc
1266	Total grid current		---	---	Ic1	---	-15	µAdc
Ⓓ 1031	Sweep-frequency vibration	No voltages; 10 G; F = 25 to 2,000 Hz See note 8	---	---	---	---	---	---
----	Vibration end points:							
1261	Electrode voltage (grid)		---	---	Ec1	-32	-46	Vdc
1266	Total grid current		---	---	Ic1	---	-15	µAdc

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NOTES:

1. When the tube is operated at 100 percent of maximum rated anode dissipation at an incoming air temperature of 25°C maximum, a minimum airflow of 5.6 cfm at sea level shall pass through the anode cooler. The static pressure drop across the anode cooler at this flow is approximately 0.26 inch of water. If the socket on Drawing 246-JAN is used, an incoming airflow of 5.6 cfm to the grid end of the socket is required. At this flow of 5.6 cfm the static pressure drop directly across the tube and socket is approximately 0.60 inch of water. This pressure drop varies with the amount of escaping air and with the shape and construction of the air director. The airflow rating applies at bias voltage less than 100 volts and frequencies less than 500 MHz. Air cooling of the tube shall be increased with increased negative grid bias, increased incoming air temperature, or increased frequency of operation, or a combination. In all cases of operation, a socket which provides forced-air cooling of the base shall be used and maximum seal and radiator temperature ratings shall not be exceeded. The airflow shall be applied before or simultaneously with electrode voltages, and may be removed simultaneously with them.

2. The filament voltage is 26.5 ±10 percent but when long life and consistent performance are factors, it is recommended that the voltage be held at ±5 percent. Maximum life may be obtained by adjusting the heater voltage in accordance with the application. A table of heater voltage versus frequency is presented as a guide.

Frequency (MHz)	Ef (Vac)
Up to 300	26.5
301 to 400	25.5
401 to 500	24.3

3. In all cases of electrical tests involving application of heater voltage, the socket on Drawing 246-JAN shall be used. An incoming airflow of 6.0 cfm maximum to the grid end of the socket is permitted.

4. An infinite baffle system as shown on figure 2, or equal, with an airflow of 5.6 cfm at sea level shall be used. The static pressure drop is measured across the tube and socket.

5. This test is to be the first test performed at the conclusion of the holding period.

6. Circuit and cavity shall be in accordance with Drawing 223-JAN.

7. This test shall be made a minimum of 30 minutes after Ef is turned off. Rated airflow shall be maintained during the 30-minute interval.

8. Test to be performed every three months, using the following sampling plan:

n1 = 4	c1 = 0	Where c2 represents the total allowable failures
n2 = 4	c2 = 1	for the first and second samples combined.

Separate samples may be used at the option of the manufacturer, and the tests shall be considered as nondestructive except in case of failure. In case of failure, after double sampling, the failing test shall become a Quality Conformance Inspection, Part 2, AQL 6.5, inspection level S3 for three consecutive successful submissions, at which time the testing may revert to the periodic-check test basis.

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NOTES: -Continued

Ⓓ. 9. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with a combined acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of Quality Conformance Inspection, part 2, inspection level S3, with an AQL of 6.5. The regular "12-calendar months" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:

Army - EL
Navy - EC
Air Force - 85

Review activities:

Air Force - 99
DLA - ES

User activities:

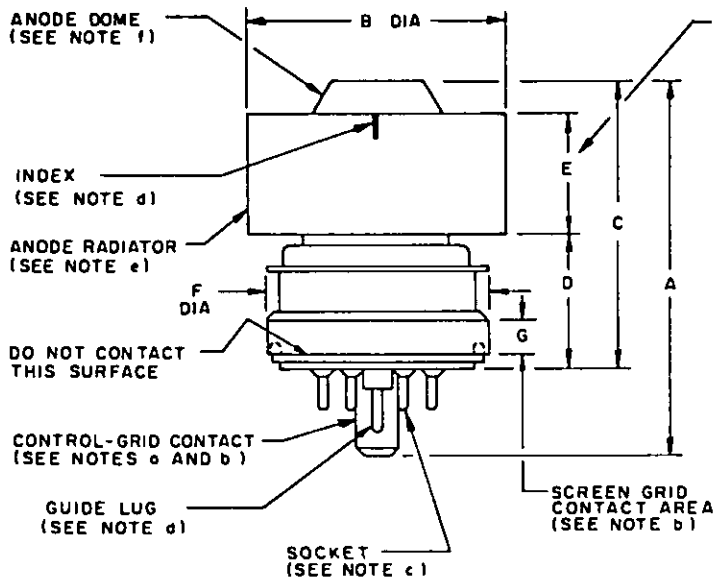
Army - WC
Navy - AS, OS, CG, SH
Air Force - 11, 19

Preparing activity:

Air Force - 85

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PIN CONNECTIONS

PIN NO.	ELEMENT
1	g 2
2	k
3	h
4	k
5	INT. CONN.
6	k
7	h
8	k
CENTER POST	g 1
RADIATOR	a
BASE RING	g 2

DIMENSIONS

LTR	INCHES		MILLIMETERS		LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 2					Q.C INSPECT., PART 3 (PERIODIC) (SEE NOTE g)				
A	2.224	2.404	56.49	61.06	B	1.610	1.640	40.89	41.66
C	1.710	1.850	43.43	46.99	D	.750	.810	19.05	20.57
					E	.710	.790	18.03	20.07
					F		1.406		35.71
					G	.187		4.75	
					H	BASE: B8-236 (SEE NOTE d)			

NOTES:

a. Pin alinement shall be checked by means of JEDEC gage GB8-3. Dimensions of control-grid contact shall be inspected by means of gages specified on Drawing 246-JAN and shall be Quality Conformance Inspection, Part 2.

b. Alinement of anode, screen grid, and control-grid contact surfaces shall be determined by means of gage specified on Drawing 168-JAN. Quality Conformance Inspection, Part 2, shall apply.

c. Air system socket shall be as specified on Drawing 246-JAN.

d. Location of guide lug of control-grid contact may be referenced by a notch or arrow on the anode radiator in the position shown.

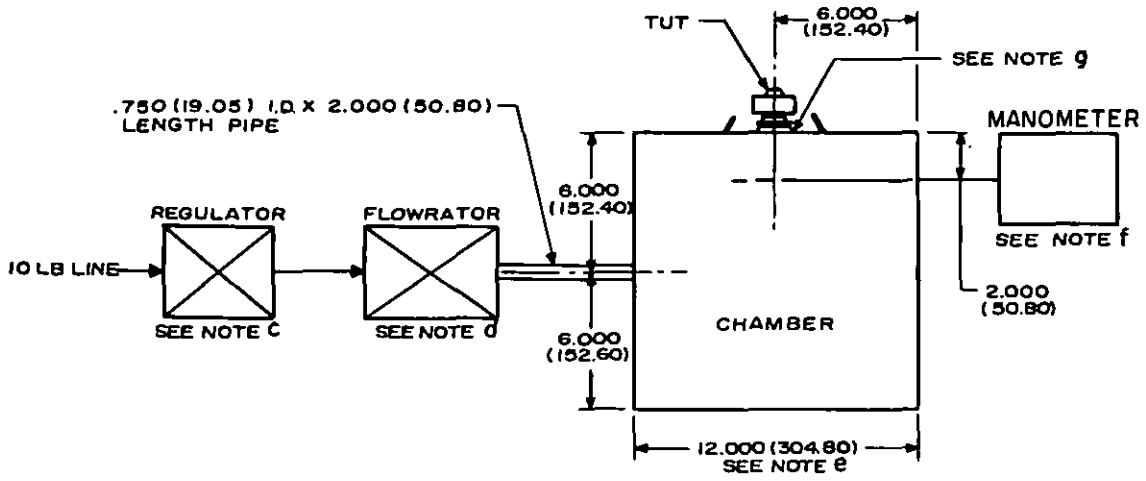
e. Anode clamping shall be confined to the anode radiator.

f. Dome contour shall be such that it will be contained within cross-hatched area of template shown on Figure 3. Optical comparator techniques are normally used for this purpose.

g. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of Quality Conformance Inspection, Part 2, inspection level S3, with an AQL of 6.5. The regular "12-calendar months" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 7609.

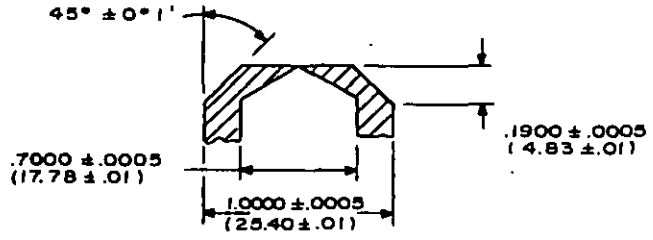
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NOTES:

a. DIMENSIONS ARE IN INCHES.
b. METRIC EQUIVALENTS ARE IN PARENTHESES.
c. FISHER-GOVERNER, PRESSURE REGULATOR MODEL 67, OR EQUAL.
d. FISHER-PORTER FLOWRATER MODEL 84-27-10/77, OR EQUAL.
e. 12 INCH (304.80 mm) CUBE INSIDE DIMENSIONS, COMPOUND SEALED, OR EQUAL.
f. F.W. DWYER MANOMETER, 0 TO 1 INCH (25.4 mm) OF WATER (FISHER SCIENTIFIC COMPANY 11-295-5 DRAFT GAGE), OR EQUAL.
g. SOCKET SPECIFIED ON DRAWING 246 JAN.

FIGURE 2. Block diagram



NOTES:

a. DIMENSIONS ARE IN INCHES.
b. METRIC EQUIVALENTS ARE IN PARENTHESES.

FIGURE 3. Contour limits of dome shape

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